Reg. No. | | | | | | | | | | | [18ASOE03]

SEVENTH SEMESTER B. TECH UNIVERSITY EXAMINATION, DECEMBER-2021

INTRODUCTION TO AEROSPACE VEHICLES AND SYSTEMS

Time: 3 Hours **Maximum Marks:70**

Instructions:

i. Answer any **FIVE** full Questions. Question number 1 and 2 are compulsory, Each question carry 14 marks.

ii. Missing data may be suitably assumed.

1.	**a.**	Explain about the parts of an airplane with its primary functions with a neat sketch.	**08**
	b.	Define the following **i.** Dihedral and Anhedral **ii.** Reynolds Number, Mach number and its classifications **iii.** Primary control and its stability axis.	**06**
2.	**a.**	What is Composite material?, and state its applications in the field of aviation.	**07**
	b.	Explain with the neat sketch about the operating principle of Turbojet Engine.	**07**
3.	**a.**	Explain in detail about the materials used in aircraft construction.	**07**
	b.	Explain in detail about Aircraft Communication System.	**07**
		OR	
4	**a.**	Explain about the operating principle of Basic Hydraulic System and its components.	**09**
	b.	List out secondary & auxiliary flight controls and describe their functions.	**05**
5.	**a.**	Explain in detail about the FADEC and EICAS with a neat sketch.	**07**
	b.	Explain about the Basic Pneumatic system and its components sketch.	**07**
		OR	
6.	**a.**	Explain in detail about the ILS and ADF with a neat sketch.	**10**
	b.	State the Design requirement for a landing gear to operate effectively.	**04**
7.	**a.**	What is meant by V –n diagram and state its importance.	**08**
	b.	Define Missile and state its classifications.	**06**
		OR	
8.	**a.**	Differentiate between Tactical and Strategic Missiles.	**04**
	b.	Define Trajectory and classify types of trajectories with a neat sketch.	**04**
	c.	Explain about the operating principle of Solid Propellant rocket with its advantages and disadvantages.	**06**